Dark Matter: Various Methods of Halo Modelling in Spiral Galaxies⋆,⋆⋆

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ABSTRACT

Observed galactic rotation curves describe the orbital velocities of stars and gas as a function of distance from the galactic centre. Under Newtonian gravity, rotational velocities are expected to decrease with radius when considering only the visible mass distribution. However, observations show that many galaxies maintain approximately flat rotation curves at large radii, providing evidence for the presence of dark matter. This study investigates three dark matter halo density profiles: cored-isothermal, Navarro–Frenk–White (NFW), and Burkert, and their ability to reproduce observed galactic rotation curves. Halo rotation curves were computed from the corresponding enclosed-mass distributions. For the cored-isothermal profile, the first-order mass-conservation ODE was solved using the fourth-order Runge–Kutta (RK4) method, and the numerical implementation was validated against the analytical solution. The resulting rotation curves were compared with observations from the Spitzer Photometry and Accurate Rotation Curves (SPARC) database. The RK4 implementation was validated against the analytical cored-isothermal solution, achieving a maximum relative velocity difference of 0.0021% and fourth-order convergence. Model fits were evaluated using reduced χ^2 and the fraction of observations within 1σ. The cored-isothermal profile provided the lowest reduced χ^2 for ten of the 18 galaxies, Burkert for seven, and NFW for one, with cored profiles generally providing better fits for dwarf and dark-matter-dominated galaxies. These results suggest that cored halo profiles better describe several galaxies in the selected sample, although the small sample, simplified baryonic treatment, and large parameter uncertainties limit the generality of this conclusion.

1. Introduction

The orbital velocities of stars and gas in spiral galaxies provide information about the distribution of mass within galaxies. Under Newtonian gravity, visible matter is expected to produce rotation curves that decrease with increasing radius. However, observations show that many spiral galaxies maintain approximately flat rotation curves at large radii, indicating the presence of additional unseen mass. This discrepancy motivates the dark matter halo model, in which galaxies are surrounded by extended distributions of matter that contribute to their gravitational fields (Navarro, Frenk and White, 1997). Different halo density profiles have been proposed, each predicting different enclosed mass distributions and rotation curves.

The enclosed mass can be obtained by deriving a first-order ordinary differential equation (ODE) from mass conservation. Solving this initial value problem gives the mass distribution (M(r)), which can then be used to determine the rotational velocity as a function of radius. This provides a direct connection between theoretical density profiles, numerical ODE methods, and observable galactic rotation curves.

This project investigates three dark matter halo profiles: cored-isothermal, Navarro–Frenk–White (NFW), and Burkert, and compares their predicted rotation curves with observations from the Spitzer Photometry and Accurate Rotation Curves (SPARC) database. Fourth-order Runge–Kutta (RK4) are used to solve the enclosed-mass ODE, with the RK4 implementation validated against the analytical cored-isothermal solution. The resulting halo models are then fitted to a selected sample of galaxies to assess their ability to reproduce observed rotation curves and identify trends in the performance of different halo profiles.

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2. Background

The orbital motion of stars and gas within a galaxy can be described by circular motion, where gravity provides the required centripetal force (Taylor, 2005). For an object of mass m orbiting at radius r,

$$\frac{GM(r)m}{r^2} = \frac{mv^2(r)}{r} \qquad \Rightarrow \qquad v(r) = \sqrt{\frac{GM(r)}{r}}, \tag{1}$$

where $M(r)$ is the enclosed mass and $v(r)$ is the circular velocity. Equation 1 therefore connects the galactic rotation curve directly to its radial mass distribution.

In this project, we focus on the dark-matter contribution to $M(r)$. Assuming spherical symmetry, the enclosed mass is determined by the halo density profile $\rho(r)$. The corresponding first-order mass-conservation IVP and its reduction from Poisson's equation, $\nabla^2\Phi = 4\pi G\rho$, are derived in Section 3.1 (Binney and Tremaine, 2008). Once $M(r)$ is obtained, Equation 1 gives the theoretical halo rotation curve.

Three dark-matter density profiles are investigated: the cored-isothermal, Navarro–Frenk–White (NFW), and Burkert profiles. Their different radial density distributions produce different enclosed-mass distributions and rotation curves, allowing their predictions to be compared with astronomical observations.

2.1. Dark Matter Halo Profiles

The isothermal sphere is a model for a self-gravitating system in which the velocity dispersion is assumed to be constant with radius (Carroll and Ostlie, 2017; Binney and Tremaine, 2008). Because this profile diverges at the galactic centre, we instead consider a cored isothermal sphere, which introduces a finite central density, ρ_0 and core radius r_c,

$$\rho_{\text{iso}}(r) = \frac{\rho_0}{1 + (r/r_c)^2}. \tag{2}$$

The Navarro–Frenk–White (NFW) profile was obtained from cosmological ΛCDM N-body simulations and is commonly used to describe the density distribution of dark matter halos (Navarro et al., 1997). The profile is given by,

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}, \tag{3}$$

where ρ_s and r_s are the characteristic density and scale radius. Unlike the cored-isothermal profile, NFW has a central density cusp meaning that its density increases significantly toward the galactic centre.

The Burkert profile (Burkert, 1995; Liu, Fu, Deng and Huang, 2005) is

$$\rho_{\text{Burkert}}(r) = \frac{\rho_0}{(1 + r/r_0)\left[1 + (r/r_0)^2\right]}, \tag{4}$$

where ρ_0 and r_0 are the central density and core radius. Unlike the NFW profile, the Burkert profile produces a finite-density core, with the density approaching ρ_0 as $r \to 0$. This cored structure makes the Burkert profile particularly useful when modelling galaxies whose observed rotation curves favour a less centrally concentrated dark matter distribution.

2.2. Mass Modelling of Spiral Galaxies

The observed rotation curve of a spiral galaxy is produced by the combined gravitational effects of its different mass components. These components include the stars, gas, and dark matter halo. Following the mass modelling approach used by the SPARC database (SPARC, n.d.), the observed rotational velocity can be decomposed as

$$V^2_{\text{obs}}(r) = V^2_{\text{gas}}(r) + V^2_{\text{disk}}(r) + V^2_{\text{bulge}}(r) + V^2_{\text{halo}}(r), \tag{5}$$

where V_{obs} is the measured rotational velocity, while V_{gas}, V_{disk},V_{bulge},and V_{halo} represent the contributions from the gaseous, stellar, and dark matter components, respectively.

3. Derivations and Numerical Methods

3.1. Reduction to the Enclosed-Mass ODE

We first derive the differential equation used to obtain the mass of the galaxy as a function of radius. Assuming spherical symmetry, we can model a galaxy as a series of thin spherical shells with radius r and thickness dr. The volume of one shell is $dV = 4\pi r^2 dr$. Since density is defined as mass per unit volume, $\rho(r) = dM/dV$, where dM is the mass contained within the shell, it follows that $dM = \rho(r)dV = 4\pi r^2 \rho(r)\, dr$. Dividing by dr gives

$$\frac{dM}{dr} = 4\pi r^2 \rho(r), \qquad M(0) = 0, \tag{6}$$

where the initial condition follows because no mass is enclosed at $r = 0$. Equation 6 therefore forms a first-order initial value problem for the enclosed mass. This equation can also be obtained by reducing the second-order gravitational-potential equation. Under spherical symmetry, Poisson's equation and the relation between gravitational potential and enclosed mass give

$$\begin{aligned} \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\Phi}{dr}\right) &= 4\pi G\rho(r), \\ r^2\frac{d\Phi}{dr} &= GM(r), \\ \frac{1}{r^2}\frac{d}{dr}\left(GM(r)\right) = 4\pi G\rho(r) \quad &\Rightarrow \quad \frac{dM}{dr} = 4\pi r^2\rho(r). \end{aligned} \tag{7}$$

Thus, introducing the enclosed mass $M(r)$ reduces the second-order gravitational-potential equation to the first-order IVP in Equation 6. For the numerical demonstration, this IVP is specialised to the cored-isothermal density profile and solved using the fourth-order Runge–Kutta method.

3.2. Cored-Isothermal Initial Value Problem

The singular isothermal sphere has a density proportional to r^{-2} and therefore diverges at the galactic centre. To introduce a finite central density, we instead use the cored-isothermal, or pseudo-isothermal, density profile $\rho(r) = \rho_0/[1 + (r/r_c)^2]$, where ρ_0 is the central density and r_c is the core radius. At the centre, $\rho(0) = \rho_0$, so the density remains finite.

Rather than solving the full isothermal Lane–Emden equation, the enclosed mass is obtained directly from the first-order ODE derived in Equation 6. Substituting the cored-isothermal density gives

$$\frac{dM}{dr} = \frac{4\pi\rho_0 r^2}{1 + \left(r/r_c\right)^2}, \qquad M(0) = 0. \tag{8}$$

This is the initial value problem solved numerically using RK4. An analytical solution is also available and is used to validate the numerical method. Integrating the same ODE gives

$$M(r) = 4\pi\rho_0 r_c^3 \left[\frac{r}{r_c} - \tan^{-1}\left(\frac{r}{r_c}\right)\right]. \tag{9}$$

Using the circular-velocity relation $V(r) = \sqrt{GM(r)/r}$ gives

$$V_{\text{halo}}(r) = \sqrt{4\pi G\rho_0 r_c^2 \left[1 - \frac{r_c}{r}\tan^{-1}\left(\frac{r}{r_c}\right)\right]}. \tag{10}$$

The analytical velocity in Equation 10 is used both in the halo fitting and as the benchmark for the numerical RK4 solution.

3.3. Fourth-Order Runge–Kutta Method

The cored-isothermal initial value problem derived above is solved numerically using the fourth-order Runge–Kutta (RK4) method. In general, RK4 provides a higher-order approximation to a first-order IVP by evaluating the derivative at four points within each step. For Equation 8, we write

$$M'(r) = f(r), \qquad f(r) = \frac{4\pi\rho_0 r^2}{1 + \left(r/r_c\right)^2}.$$

Since f depends explicitly on r but not on M, the RK4 stages for a radial step size h reduce to

$$k_1 = f(r_n), \qquad k_2 = f\left(r_n + \frac{h}{2}\right), \qquad k_3 = f\left(r_n + \frac{h}{2}\right), \qquad k_4 = f(r_n + h). \tag{11}$$

and the enclosed mass is advanced according to

$$M_{n+1} = M_n + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right). \tag{12}$$

Beginning with $M(0) = 0$, RK4 is applied successively across the radial grid to obtain $M(r)$. The numerical mass is then converted to rotational velocity using $V(r) = \sqrt{GM(r)/r}$ and compared with the analytical cored-isothermal solution to assess the accuracy and convergence of the method.

4. Methods

Table 1
Properties of the galaxies selected from the SPARC database for dark matter halo modeling.

Galaxy	$N_{\rm pts}$	Q	$V_{\rm flat}$ (km/s)	Regime
NGC 7331	36	1	239.0	Massive spiral
NGC 2903	34	1	184.6	Massive spiral, baryonic dominated
NGC 3198	43	1	150.1	Intermediate spiral
NGC 3521	41	1	213.7	Massive spiral
NGC 5055	28	1	179.0	Massive spiral
NGC 2403	73	1	131.2	Intermediate spiral
NGC 6503	31	1	116.3	Intermediate spiral, LSB, DM dominated
NGC 4559	32	1	121.2	Late spiral
NGC 5585	24	1	90.3	Intermediate spiral, LSB, DM dominated disk
NGC 1003	36	1	109.8	Intermediate spiral, LSB, baryonic dominated
UGC 03546	30	1	196.9	Early spiral
UGC 00128	22	1	129.3	LSB
DDO 154	(2.42±	2	47.0	Dwarf, DM dominated
NGC 3741	21	1	50.1	Dwarf
DDO 161	31	1	66.3	Dwarf
IC 2574	34	1	66.4	Dwarf
UGC 05721	23	1	79.7	Dwarf
UGCA 442	8	1	56.4	Dwarf

$N_{\rm pts}$ is the number of observed rotation-curve data points.

The galaxies used in this study were pulled from the SPARC (Spitzer Photometry and Accurate Rotation Curves) database. It contains, in its entirety, data on 175 spiral and dwarf galaxies. Since the goal of this work is to compare different dark matter halo models, a sample was chosen from the complete selection of galaxies.

There were several fundamental criteria used in selection of the final sample of galaxies. The SPARC database assigns a quality flag to each galaxy dataset, primarily, preference was given to data with high quality. Data chosen in this study

was restricted to quality of Q = 1, and excluded data of classes Q = 2, and Q = 3, with the exception of galaxy DDO154 because it serves as a highly dark matter dominated dwarf galaxy which the other sample dwarves don't demonstrate. Additionally, galaxies with a large number of measured rotation points over a large radial range were prioritised. Most importantly, in order to determine the efficacy of a model in fitting to galaxy rotation curves, morphologically variable galaxies were necessary. The final sample includes galaxies ranging the progression of spirals (late, intermediate, massive, early), dwarf galaxies, low-surface brightness, baryonic contribution dominated, and dark matter dominated. Another consideration was narrowing the size of V_{flat} to filter galaxies. A large V_{flat} is indicative of a large quantity of enclosed mass in the inner bulge and disk of a galaxy, whereas a lower V_{flat} is indicative of less enclosed mass. A variety of V_{flat} values were included in order to observe the interactions of different dynamical regimes with the rotation curve fits.

A smaller sample of 18 selected galaxies was used to permit a more detailed study of the individual fits and the resulting halo parameters within time constraints. The initial galaxy selection was informed by an AI-assisted screening of the 175 SPARC galaxies to identify suitable candidates based on the stated criteria. An emphasis was placed on using a variety of regimes of galaxy to preserve flexibility and representation across a range of galaxy regimes. The resulting candidates were then independently examined using the SPARC rotation-curve data before the final sample was selected.

4.1. Rotation Curve Modelling and Parameter Fitting

Burkert, NFW, and cored-isothermal halo profiles were fitted to each selected galaxy where each halo profile's total rotational velocity was calculated by combining the SPARC baryonic contributions with the dark-matter halo contributions. The total rotational velocity, (5), was directly implemented in the computational model, where each of the parameters (gas, disk, and bulge) were taken from the SPARC data. The halo velocities were then evaluated from the final selection of the eighteen galaxy profiles. The two free halo parameters for each model were optimised by using the standard least–squares routine from scipy.optimize, and the function was defined using normalised residuals from uncertainties $R_i = \frac{V_{\mathrm{model},i} - V_{\mathrm{obs},i}}{\sigma_{V,i}}$. This was done so that the reported observations with smaller reported uncertainties contributed more strongly to the fit. Additionally, both halo parameters were restricted to only positive values to avoid mathematical errors.

Initial estimates for the order of magnitudes were input to initialise the first steps of the nonlinear optimization. For the Burkert and cored-isothermal fits, the initial density and core radius was set to 107 and 5 kpc, while the NFW fits were set to 106 and 10 kpc. However, these starting values were not imposed on the final fitted solutions. Further, the approximate parameter uncertainties were calculated from the Jacobian matrix returned by the least–squares optimiser, and the covariance matrix was approximated by $(\mathbf{J}^T\mathbf{J})^{-1}$, with parameter uncertainties estimated from the square roots of its diagonal elements.

For model comparisons, the reduced χ^2 statistic was used, with $N - 2$ degrees of freedom for N measured rotation-curve points and two fitted halo parameters. Lower values of χ^2_r indicate a closer agreement between theoretical and observed rotation curves. Residual plots were additionally used to identify errors and the difference from observations that may not be apparent from simply calculating a goodness-of-fit statistic. These quantities were calculated for each of the three dark-matter halo models.

4.2. Numerical ODE Demonstration using RK4

A numerical solution was used for the mass-conservation ODE of the cored isothermal profile using NGC3741 as a representative case. Fitted values of ρ_0 (central density of the dark matter halo) and r_c (core radius of the dark matter halo) were substituted in the derived equation 8, with M(0) = 0. This initial value problem was solved numerically using the fourth order Runge-Kutta method (RK4). The resulting enclosed mass was then converted to a rotational velocity using the mass value found from the numerical solution. Plots were produced to compare the analytical cored-isothermal velocity with the numerical solution. Although any of the galaxies could have been used for the comparative analysis, NGC3741 was selected because it exhibited a strong correlation between the model and the observed rotation curve, with a low χ^2 value of 0.56. It was also of particular interest as a dwarf galaxy, for dark halo modelling is especially relevant to the core-cusp problem agreement. All computations were performed in Python using Jupyter

Table 2
RK4 convergence test for the cored-isothermal enclosed-mass ODE using NGC3741.

N	h (kpc)	Absolute mass error ($M_\odot$)	Error ratio	Observed order
20	0.35	13.44	15.98	3.998
40	0.175	0.8409	15.99	4
80	0.0875	0.05258	15.99	3.999
160	0.04375	0.003289	–	–

Notebook. NumPy was used for numerical operations and data handling, SciPy for nonlinear least-square optimisation, and Matplotlib for visualisation.

5. Results and Analysis

The resulting rotation-curve fits for the selected galaxies are shown in Figure 2 in the Appendix.

5.1. Analysis RK4 Numerical ODE Demonstration

The numerical result was validated against the analytical enclosed-mass solution by using the RK4 methods. Figure 1 describes the nearly indistinguishable velocity curves between the two solutions. Across the calculated radial range, the maximum absolute velocity difference was approximately $3 \times 10^{-5} \frac{km}{s}$, corresponding to a maximum relative difference of approximately 0.0021%. It can also be noted that at 7 kpc, the RK4 and analytical enclosed masses were both around $3.89 \times 10^9 M_\odot$. These values demonstrate that the numerical error was negligible relative to the physical scale of the system. A convergence test, detailed in Table 2 further confirmed the expected behaviour of RK4. Successive halving of the radial step size reduced the numerical mass error by a factor of 16, giving an observed convergence order of 4. This is consistent with the theoretical global error, $O(h^4)$, of the fourth-order Runge-Kutta method, providing numerical validation of the implementation.

The behaviour of the resulting curve in Figure 1 follows directly from the cored-isothermal density distribution. Within the core, the density approaches a constant value, producing $M(r) \propto r^3$ and therefore $V(r) \propto r$. At larger radii, the density approaches $\rho(r) \propto r^{-2}$, causing the enclosed mass to increase almost linearly with radius. As a result, $V(r) = \sqrt{\frac{GM(r)}{r}}$ approaches a constant value rather than undergoing a Keplerian decline. The numerical RK4 solution therefore reproduces both the analytical solution and the characteristic physical flattening expected from an isothermal dark matter halo. Altogether, the RK4 analysis demonstrated that the numerical method accurately reproduces the known solution of the underlying first-order initial value problem.

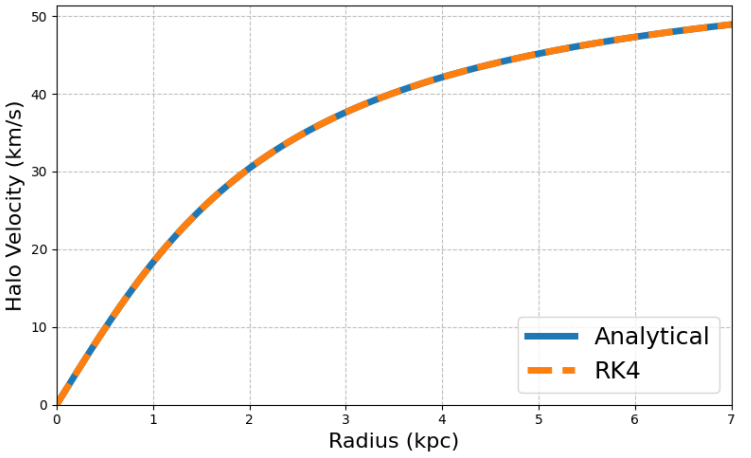


Figure 1: Comparison of the analytical and fourth-order Runge–Kutta (RK4) solutions for the cored-isothermal halo of NGC3741. The nearly overlapping rotation curves demonstrate the accuracy of the numerical solution.

5.2. Analysis of Rotation Curve Fits

The reduced χ_r^2 values in Table 4 show substantial variation in the quality of dark matter halo fits between galaxies. In general, all three fits produced comparable fits for several of the spiral galaxies, whereas larger differences were observed among the fits to dwarf galaxies.

Four massive spiral galaxies were studied: NGC7331, NGC2903, NGC3521 and NGC5055. The fits were generally poor for these galaxies, produced χ_r^2 values significantly greater than 1. In particular, NGC5055 has χ_r^2 all greater than 3000, indicating that none of the halo fits provide an accurate description of its observed rotational velocity. Each fit produced similar values in this case, indicating that the poor fit is not necessarily a consequence of halo model choice. Each massive spiral galaxy produced the smallest value of χ_r^2 under the isothermal sphere model - however none approached unity, and each fit performed remarkably similarly across these galaxies.

The dwarf galaxies (DDO154, NGC3741, DDO161, IC2574, UGC05721, UGCA442) typically showed a more pronounced difference between models. The NFW model provided much larger χ_r^2 values. The difference is particularly large for IC 2574 where χ_r^2 was 35.28 for the NFW model and approached 2 in both the Burkert and Isothermal sphere models. Similarly, this applies to DDO161. These results suggest that, for several of the dwarf galaxies in the sample, the inner density structure represented by the Burkert and isothermal profiles provides a substantially better description of the observed rotation curves than the NFW profile.

The LSB (Low surface brightness, NGC6503, NGC5585, NGC1003, and UGC00128) galaxies generally show a preference against NFW relative to the cored Burkert and isothermal profiles, although this trend is not universal. NFW is the worst fitting model for both NGC6503, and NGC5585, but it performs comparably to both other models in NGC1003 and outperforms Burkert in UGC00128. The LSB galaxies show a tendency toward improved fits from Burkert and cored–isothermal profiles relative to the NFW profile.

The dark-matter-dominated subsample (DDO154, NGC6503, and NGC5585) exhibits a consistent preference for cored halo profiles. For all three DM-dominated galaxies, the NFW model produces a larger reduced χ_r^2 than both the Burkert and isothermal models. The difference is particularly pronounced for DDO154 and NGC5585, suggesting that the observed inner rotation curves of these galaxies are better described by models with a finite-density core than by the NFW profile

Within the selected sample, The percentage of points lying within 1 σ of the observations provides a supplementary measure of fit quality. The dwarf galaxies typically have much larger overlaps than the spiral galaxies. For example, UGCA442 has 87.5% of its points within 1σ for the Burkert model, compared with 50.0% for NFW. Similarly, UGC05721 has an 87.0% overlap for Burkert and 73.6% for the isothermal model. This is consistent with the χ_r^2 results and provides additional evidence that the Burkert profile performs well for several of the dwarf galaxies (Burkert, 1995).

The two statistics, however, do not always identify the same model as the best fit. For example, UGC 00128 has the lowest reduced chi-squared for the isothermal profile ($\chi_r^2 = 3.35$), closely followed by NFW ($\chi_r^2 = 3.42$), while the NFW model has a larger fraction of points within 1σ than Burkert. The percentage of points within 1σ should be interpreted as a supplementary diagnostic rather than as a replacement for the χ_r^2 statistic, since it does not account for how far points lie from the model once they fall outside the 1σ interval.

The uncertainties on the fitted halo parameters in Table 3 vary substantially between galaxies and between halo models. In several cases, particularly for the NFW scale radius, the formal uncertainties are comparable to or substantially larger than the best-fitting values. This indicates that the available rotation-curve data do not strongly constrain the individual halo parameters, despite the models producing visually similar rotation curves.

Although the best-fitting models provide estimates of the halo density and scale radii, the formal parameter uncertainties are large for several galaxies, especially in the case of the NFW scale radius. In some cases the uncertainty exceeds the best-fitting value by more than an order of magnitude, indicating strong parameter issues. Additionally, several fits approach the imposed lower parameter bounds, for which the covariance-based uncertainty estimates are not reliable. The fitted parameters should be interpreted cautiously.

Overall, the results suggest that there is no one model that provides the best fit for every galaxy. The Burkert and Isothermal sphere profiles generally perform similarly, whereas the NFW model provides poorer fits (especially in the context of the dwarf galaxies). Some galaxies show little distinction between the models, especially in rotation curve

data that does not closely constrain the rotation curve behaviour. The large χ_r^2 values for several galaxies indicate discrepancies between the models and observational data cannot necessarily be resolved by changing the halo profile alone. These discrepancies may instead reflect limitations in the assumed baryonic contributions, uncertainties in the observational data, or the simplified two-parameter halo models used in this analysis.

6. Discussion

This project successfully developed a Python framework for modeling dark-matter rotation curves with the SPARC database. The analysis allowed Burkert, NFW, and isothermal halo profiles to be fitted consistently to the observed rotation curves while accounting for observational velocity uncertainties. Comparing the models using χ_r^2, normalized residuals, and the fraction of points reproduced within 1σ provided multiple measures of fit quality. The analysis also demonstrated that the choice of galaxy sample can affect the conclusions drawn from a halo-model comparison. In particular, the selected dark-matter-dominated and low-surface-brightness galaxies generally showed better fits from the Burkert and cored isothermal profiles than from the NFW profile, although the sample size was too small to establish this as a general property of these galaxy populations.

Several limitations were, however, encountered in this study. The sample only selected 18 of 175 galaxies provided in the SPARC database, meaning that the results may be affected by selection bias and are not necessarily representative of the full database. Additionally, several fitted coefficients were poorly constrained. In particular, many of the NFW scale radii had massive uncertainties, indicating strong degeneracies between the halo parameters. Several of the galaxies provided no strong preferential fit to any of the three dark-matter halo modelling methods used based on the χ_r^2, indicating that perhaps no galaxy type constrains consistently well to any of these models, or that simply a different quality evaluation should be applied.

The analysis of the data also relied on simplifying assumptions that aren't necessarily physically accurate. The baryonic and dark-matter contributions were combined according to equation 5, while the SPARC baryonic contributions were treated as fixed. The halo was additionally assumed to be adequately described by one of the three analytic density profiles considered. These assumptions neglect uncertainties in quantities such as observational or modeling uncertainties that may influence the inferred halo parameters given by the SPARC database. Additionally, we assumed a symmetrical sphere model for the galaxies, where this isn't necessarily physically accurate.

A natural extension of this study would be to apply a consistent, quantitative selection criterion to the full SPARC sample (or to simply extend to the full SPARC sample) and investigate whether the preference for cored profiles persists across different galaxy populations. The parameter-fitting procedure could also be improved using a Monte Carlo approach, which would provide improved parameter uncertainties and allow correlations between halo parameters to be examined directly. The study could also be extended to study dark matter in not only spiral and dwarf galaxies, but additionally elliptical and other-type galaxies using a more comprehensive database. Perhaps even extended to fully evaluate the cusp/core problem encountered in dwarf galaxy rotation curve fitting.

7. Conclusion

This study investigated several dark matter halo profiles alongside the derived first-order ordinary differential equation describing the enclosed mass, and compared the fits of these profiles using numerical methods to obtain corresponding galactic rotations curves. Three density profiles: cored-isothermal, Navarro–Frenk–White (NFW), and Burkert were considered and compared with SPARC observations. The RK4 implementation was validated using the cored-isothermal profile, reproducing the analytical rotation curve with a maximum absolute velocity difference of approximately $3 \times 10^{-5}, \mathrm{km}, \mathrm{s}^{-1}$ and a maximum relative difference of 0.0021%. Halving the radial step size reduced the numerical error by a factor of approximately 16, giving an observed convergence order of four, consistent with the expected $O\left(h^4\right)$ global error of RK4. This validates the numerical implementation and demonstrates the usefulness of numerical integration for halo profiles without convenient analytical mass solutions.

The halo models were then fitted to a selected sample of 18 SPARC galaxies. No single profile provided the best fit across the entire sample: the cored-isothermal model produced the lowest reduced χ^2 for ten galaxies, Burkert

for seven, and NFW for one. Dwarf and dark-matter-dominated galaxies showed the clearest preference for cored profiles, with Burkert and cored-isothermal models generally outperforming NFW. However, several massive spiral galaxies were poorly fitted by all models, indicating that discrepancies cannot necessarily be resolved through the halo profile alone. The results are further limited by the small, selected sample, spherical-symmetry assumption, simplified baryonic treatment, and large uncertainties in some fitted parameters. Thus, while the analysis does not establish a universal preferred halo profile, it demonstrates the usefulness of numerical ODE methods for connecting theoretical mass distributions with observed galactic dynamics and identifies a notable preference for cored profiles among several dwarf and dark-matter-dominated galaxies.

A. Link to Repository

https://github.com/MamonaK/Dark_Matter_Halo_Modelling.git

B. Tables

Table 3
Best-fit dark matter halo parameters for the selected SPARC galaxies.

Galaxy	ρ_{burkert}	ρ_{NFW}	$\rho_{\text{isothermal}}$	r_{burkert}	r_{NFW}	$r_{\text{isothermal}}$
DDO154	$2.42 \times 10^{7} \pm 1.9 \times 10^{-9}$	$8.6 \times 10^{5} \pm 6.6 \times 10^{5}$	$2.18 \times 10^{7} \pm 1.0 \times 10^{-9}$	2.84 ± 0.02	23 ± 15	1.74 ± 0.02
DDO161	$4.91 \times 10^{6} \pm 2.6 \times 10^{5}$	$1.9 \times 10^{4} \pm 6.3 \times 10^{4}$	$4.29 \times 10^{6} \pm 2.7 \times 10^{5}$	8.68 ± 0.37	559 ± 1840	5.46 ± 0.29
IC2574	$2.28 \times 10^{6} \pm 2.1 \times 10^{5}$	$8.7 \times 10^{3} \pm 1.8 \times 10^{5}$	$2.17 \times 10^{6} \pm 1.3 \times 10^{5}$	45 ± 33	793 ± 16500	20 ± 9
NGC1003	$5.09 \times 10^{6} \pm 3.9 \times 10^{5}$	$2.53 \times 10^{5} \pm 7.7 \times 10^{4}$	$5.01 \times 10^{6} \pm 4.7 \times 10^{5}$	13.79 ± 0.74	83 ± 20	7.78 ± 0.54
NGC2403	$1.84 \times 10^{7} \pm 1.2 \times 10^{6}$	$4.10 \times 10^{5} \pm 2.4 \times 10^{5}$	$1.66 \times 10^{7} \pm 1.3 \times 10^{6}$	8.03 ± 0.41	97 ± 51	4.90 ± 0.31
NGC2903	$1.5 \times 10^{6} \pm 7.9 \times 10^{5}$	$2.4 \times 10^{3} \pm 1.8 \times 10^{5}$	$1.5 \times 10^{6} \pm 1.2 \times 10^{6}$	$1.2 \times 10^{3} \pm 3.6 \times 10^{4}$	$5.0 \times 10^{3} \pm 3.7 \times 10^{5}$	$1.2 \times 10^{3} \pm 2.5 \times 10^{6}$
NGC3198	$4.00 \times 10^{6} \pm 5.1 \times 10^{5}$	$3.6 \times 10^{4} \pm 7.0 \times 10^{4}$	$3.44 \times 10^{6} \pm 5.1 \times 10^{5}$	19.4 ± 1.7	493 ± 879	12.2 ± 1.4
NGC3521	$2.1 \times 10^{4} \pm 1.8 \times 10^{6}$	$7.43 \times 10^{2} \pm 0$	$3.6 \times 10^{5} \pm 1.8 \times 10^{6}$	$3.4 \times 10^{3} \pm 1.5 \times 10^{5}$	0.02 ± 0	$2.2 \times 10^{5} \pm 1.1 \times 10^{5}$
NGC3741	$2.34 \times 10^{7} \pm 4.9 \times 10^{-9}$	$4.3 \times 10^{5} \pm 2.1 \times 10^{5}$	$2.23 \times 10^{7} \pm 1.8 \times 10^{-9}$	2.98 ± 0.05	41 ± 18	1.72 ± 0.03
NGC4559	$2.4 \times 10^{6} \pm 7.1 \times 10^{5}$	$4.1 \times 10^{3} \pm 1.3 \times 10^{5}$	$2.1 \times 10^{6} \pm 5.5 \times 10^{5}$	28 ± 14	$3.4 \times 10^{3} \pm 1.1 \times 10^{5}$	18 ± 8
NGC5055	$1.67 \times 10^{3} \pm 0$	$8.23 \times 10^{-2} \pm 0$	$3.9 \times 10^{5} \pm 2.4 \times 10^{5}$	0 ± 0	0 ± 0	$1.6 \times 10^{4} \pm 3.7 \times 10^{6}$
NGC5585	$7.3 \times 10^{6} \pm 3.1 \times 10^{6}$	$2.0 \times 10^{4} \pm 6.2 \times 10^{4}$	$6.5 \times 10^{6} \pm 2.5 \times 10^{6}$	13 ± 7	980 ± 2960	7.9 ± 4.0
NGC6503	$1.08 \times 10^{7} \pm 2.1 \times 10^{6}$	$6.9 \times 10^{5} \pm 5.2 \times 10^{5}$	$1.00 \times 10^{7} \pm 2.6 \times 10^{6}$	9.2 ± 1.2	46 ± 26	5.46 ± 1.05
NGC7331	$2.07 \times 10^{4} \pm 0$	4.56 ± 0	$3.1 \times 10^{5} \pm 3.7 \times 10^{5}$	0 ± 0	0 ± 0	$4.5 \times 10^{3} \pm 6.4 \times 10^{6}$
UGC00128	$1.23 \times 10^{7} \pm 1.1 \times 10^{6}$	$2.55 \times 10^{6} \pm 2.6 \times 10^{5}$	$2.39 \times 10^{7} \pm 1.3 \times 10^{-9}$	9.58 ± 0.48	21.36 ± 1.26	3.49 ± 0.02
UGC03546	$1.3 \times 10^{6} \pm 1.9 \times 10^{5}$	$2.3 \times 10^{3} \pm 2.4 \times 10^{5}$	$1.4 \times 10^{6} \pm 2.0 \times 10^{5}$	$7.7 \times 10^{3} \pm 9.5 \times 10^{2}$	$6.0 \times 10^{3} \pm 6.3 \times 10^{5}$	$5.1 \times 10^{4} \pm 2.5 \times 10^{4}$
UGC05721	$2.79 \times 10^{8} \pm 4.2 \times 10^{-11}$	$3.69 \times 10^{7} \pm 5.1 \times 10^{-9}$	$3.30 \times 10^{8} \pm 1.4 \times 10^{-11}$	1.29 ± 0.02	3.62 ± 0.09	0.61 ± 0.01
UGCA442	$2.89 \times 10^{7} \pm 2.1 \times 10^{-9}$	$7.8 \times 10^{5} \pm 6.3 \times 10^{5}$	$2.72 \times 10^{7} \pm 1.0 \times 10^{-9}$	2.98 ± 0.03	28 ± 19	1.73 ± 0.02

Values of ρ in units of $M_{\odot}\,\text{kpc}^{-3}$
Values of r in units of kpc

Table 4
Goodness-of-fit statistics for the dark matter halo models.

Galaxy	$\chi^2_{r,\mathrm{Burkert}}$	$\chi^2_{r,\mathrm{NFW}}$	$\chi^2_{r,\mathrm{isothermal}}$	$\%_{\mathrm{Burkert}}$	$\%_{\mathrm{NFW}}$	$\%_{\mathrm{isothermal}}$
NGC 7331	171.66	171.66	**163.70**	2.8	2.8	5.6
NGC 2903	186.81	211.80	**186.71**	5.9	0	5.9
NGC 3198	**7.45**	15.39	7.84	41.9	20.9	32.6
NGC 3521	30.89	30.91	**30.53**	2.4	2.4	2.4
NGC 5055	3689.40	3689.40	**3226.39**	0	0	7.1
NGC 2403	21.44	40.65	**21.25**	42.5	23.3	41.1
NGC 6503	**17.72**	24.19	18.49	58.1	35.5	38.7
NGC 4559	2.77	4.15	**2.72**	31.2	34.4	31.2
NGC 5585	15.46	33.39	**15.11**	20.8	20.8	20.8
NGC 1003	2.72	3.02	**2.46**	50.0	50.0	44.4
UGC 03546	26.77	33.82	**26.42**	10.0	0	10.0
UGC 00128	9.33	3.42	**3.35**	36.4	45.5	54.5
DDO 154	**2.89**	16.67	4.27	50.0	8.3	41.7
NGC 3741	**0.62**	0.49	0.56	81.0	85.7	76.2
DDO 161	0.57	2.12	**0.62**	77.4	38.7	74.2
IC 2574	**2.36**	35.28	2.37	73.5	38.2	73.5
UGC 05721	0.44	1.31	**0.85**	87.0	56.5	73.6
UGCA 442	0.62	2.01	**0.80**	87.5	50.0	75.0

The percentage values represent the fraction of observed rotation-curve points for which the model prediction lies within 1σ of the observed velocity.

C. Complete Rotation Curve Fits

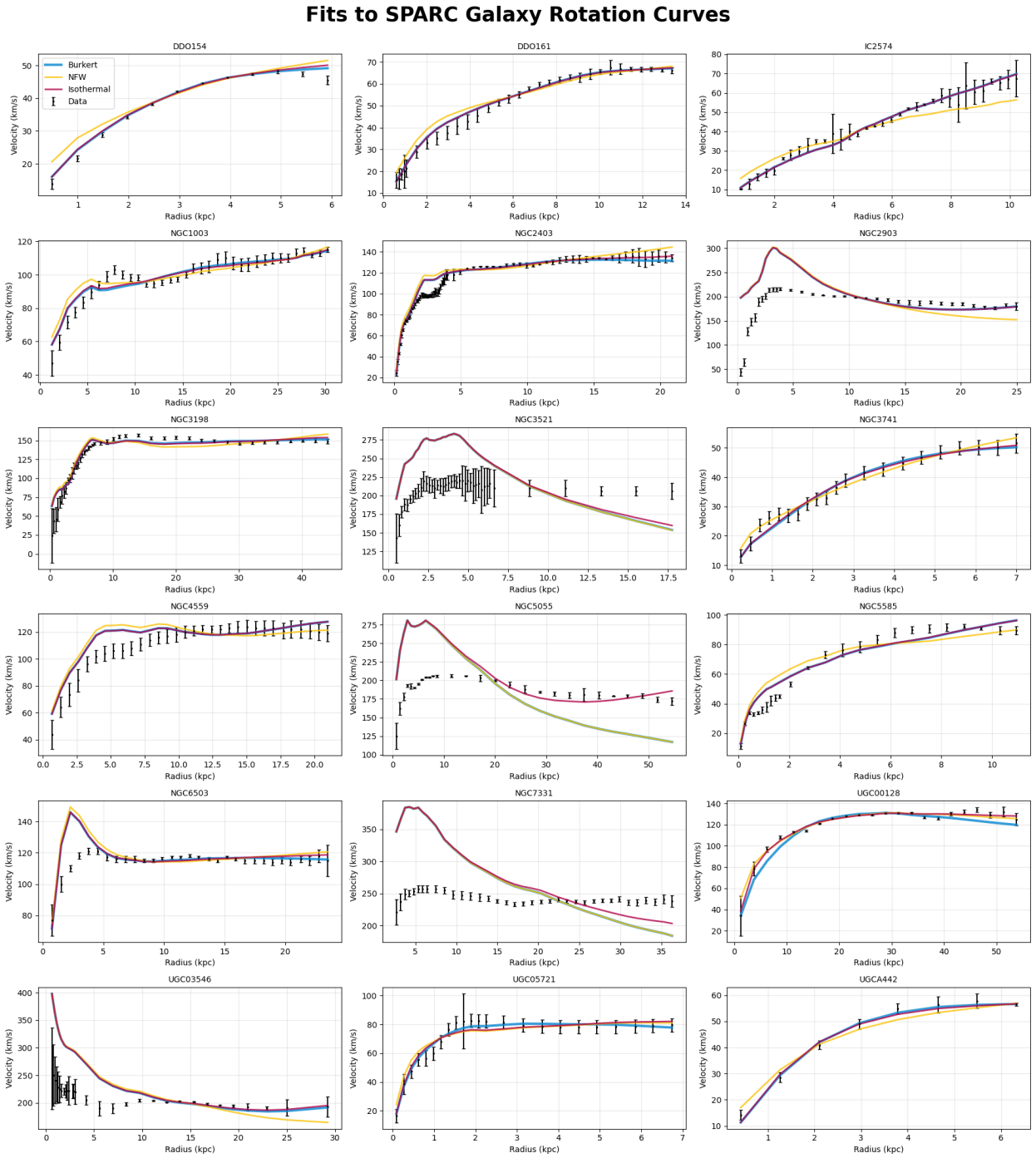


Figure 2: Observed rotation curves and best-fitting dark matter halo models for the selected galaxies. The observed rotational velocities are compared with the cored-isothermal, Burkert, and Navarro–Frenk–White (NFW) profiles.

References

Allaert, F., Gentile, G., Baes, M., 2017. Testing baryon-induced core formation in ΛCDM: A comparison of the dc14 and corenfw dark matter halo models on galaxy rotation curves. Astronomy & Astrophysics 605, A55. doi:10.1051/0004-6361/201730402.

Binney, J., Tremaine, S., 2008. Galactic Dynamics. 2 ed., Princeton University Press.

de Blok, W.J.G., 2009. The core-cusp problem. Advances in Astronomy 2010. URL: http://dx.doi.org/10.1155/2010/789293, doi:10.1155/2010/789293.

Burkert, A., 1995. The structure of dark matter halos in dwarf galaxies. The Astrophysical Journal Letters 447, L25–L28. doi:10.1086/309560.

Carroll, B.W., Ostlie, D.A., 2017. An Introduction to Modern Astrophysics. 2 ed., Cambridge University Press. doi:10.1017/9781108380980.

Lelli, F., McGaugh, S.S., Schombert, J.M., 2016. Sparc: Mass models for 175 disk galaxies with spitzer photometry and accurate rotation curves. The Astronomical Journal 152, 157. doi:10.3847/0004-6256/152/6/157.

Liu, W., Fu, Y., Deng, Z., Huang, J., 2005. An equilibrium dark matter halo with the burkert profile. arXiv doi:10.48550/arXiv.astro-ph/0505579, arXiv:astro-ph/0505579.

McGaugh, S.S., Lelli, F., Schombert, J.M., 2016. Radial acceleration relation in rotationally supported galaxies. Physical Review Letters 117, 201101. doi:10.1103/PhysRevLett.117.201101.

Navarro, J.F., Frenk, C.S., White, S.D.M., 1997. A universal density profile from hierarchical clustering. The Astrophysical Journal 490, 493–508. doi:10.1086/304888.

Salucci, P., Burkert, A., 2000. Dark matter scaling relations. arXiv doi:10.48550/arXiv.astro-ph/0004397, arXiv:astro-ph/0004397.

Sanders, R.H., 2010. Direct evidence: Extended rotation curves of spiral galaxies, in: The Dark Matter Problem: A Historical Perspective. Cambridge University Press, Cambridge, pp. 38–56.

SPARC, n.d. Spitzer photometry & accurate rotation curves. URL: https://astroweb.case.edu/SPARC/. accessed August 9, 2026.

Taylor, J.R., 2005. Classical Mechanics. University Science Books.

Yaakub, A.R., Evans, D.J., 1999. A fourth order runge–kutta rk(4,4) method with error control. International Journal of Computer Mathematics 71, 383–411. URL: https://doi.org/10.1080/00207169908804817, doi:10.1080/00207169908804817.